Artifact 2 — SIEM Detection Lab

Detecting an SSH Brute-Force Attack with Wazuh

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Artifact	2 of 5 — SIEM & Detection	**Platform**	Wazuh 4.14.7

1. Objective

This artifact demonstrates that I can stand up a Security Information and Event Management (SIEM) platform, connect it to a monitored host, and detect a real attack against that host from the collected logs. A SIEM is a central system that gathers log data from many machines and raises alerts when the activity matches known attack patterns. The specific scenario proven here is the detection of an **SSH brute-force attack** — an attacker repeatedly guessing the password for the remote-login service (SSH) until they either get in or are noticed.

2. Lab Environment

The lab runs two virtual machines on an isolated internal network so that the attack traffic never touches a real network:

SIEM platform	Wazuh 4.14.7 (manager + dashboard), cluster node **node01**
Monitored host (agent)	**ubuntu-victim** — Ubuntu 24.04.4 LTS, agent ID 001, IP 10.0.2.4
Agent group	default
Attack simulated	Repeated failed SSH logins (brute-force) against the monitored host

3. Step 1 — Enroll the Agent and Confirm It Is Reporting

Before any attack can be detected, the monitored host must be sending its logs to the SIEM. This is done by installing the **Wazuh agent** — a small program on the host that forwards log events to the Wazuh manager. The screenshot below confirms the agent enrolled successfully and is in an **active** state, meaning it has a live connection to the manager and its logs are flowing in.

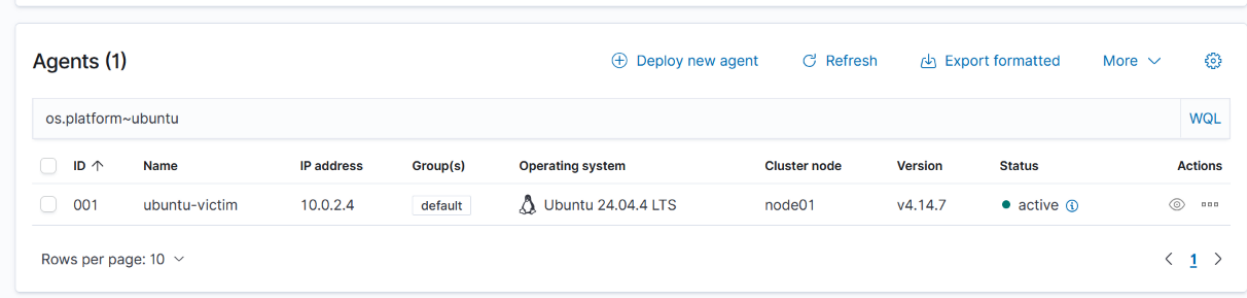
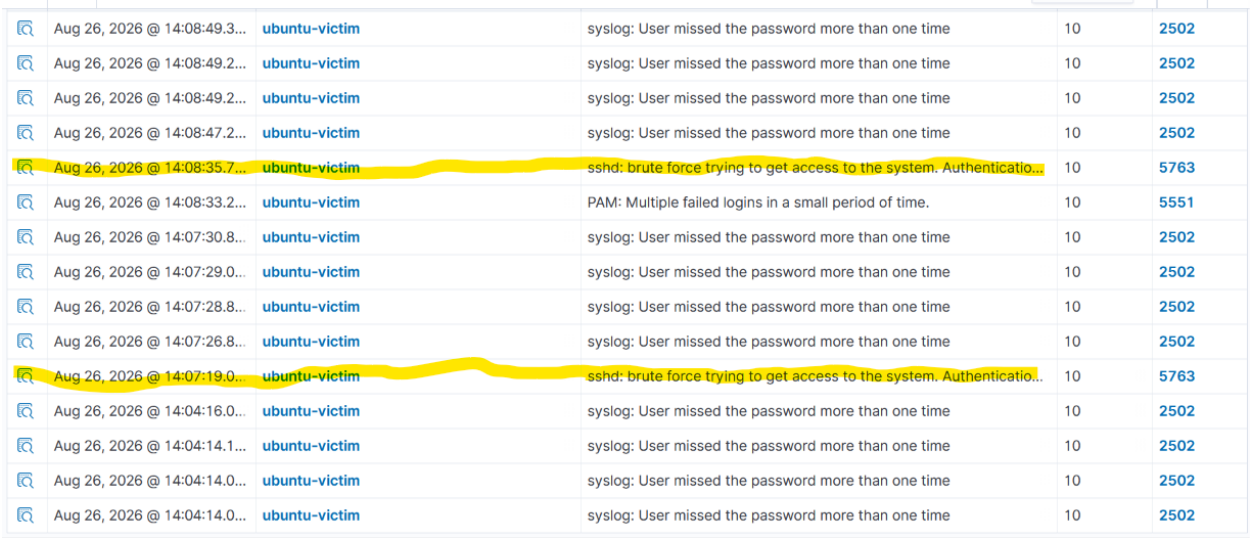


Figure 1. Wazuh Agents view: host **ubuntu-victim** (ID 001, 10.0.2.4, Ubuntu 24.04.4 LTS) shows status **active** on manager node01 — the prerequisite for any detection. Evidence: A2-S06-wazuh-agent-active-dashboard.png

4. Step 2 — Trigger the Attack and Detect It

With logs flowing, I generated repeated failed SSH login attempts against ubuntu-victim to imitate an attacker guessing passwords. Wazuh reads each failed-login line from the host's authentication log and matches it against its built-in rule set. When enough failures happen in a short window, Wazuh escalates from low-level noise to a high-severity brute-force alert. The detection chain that fired is shown below.

2502	User missed the password (single failed login)	Low-level signal — one failure is normal noise
5712	PAM: multiple failed logins in a small period of time	Correlation — groups many failures together
5763	sshd: brute-force trying to get access to the system	**The detection** — the alert that confirms an attack



Aug 26, 2026 @ 14:08:49.3...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:08:49.2...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:08:49.2...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:08:47.2...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:08:35.7...	ubuntu-victim	sshd: brute force trying to get access to the system. Authenticatio...	10	5763
Aug 26, 2026 @ 14:08:33.2...	ubuntu-victim	PAM: Multiple failed logins in a small period of time.	10	5551
Aug 26, 2026 @ 14:07:30.8...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:07:29.0...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:07:28.8...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:07:26.8...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:07:19.0...	ubuntu-victim	sshd: brute force trying to get access to the system. Authenticatio...	10	5763
Aug 26, 2026 @ 14:04:16.0...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:04:14.1...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:04:14.0...	ubuntu-victim	syslog: User missed the password more than one time	10	2502
Aug 26, 2026 @ 14:04:14.0...	ubuntu-victim	syslog: User missed the password more than one time	10	2502

Figure 2. Wazuh alerts from agent ubuntu-victim: individual failed-login events (rule 2502) are correlated into a multiple-failure alert (rule 5712) and escalated to the brute-force detection (rule **5763**). Evidence: A2-S07-ssh-bruteforce-detection-5763.png

5. How the Detection Works

The value of this setup is **correlation**. A single wrong password (rule 2502) is not suspicious — people mistype passwords constantly. Wazuh only raises the alarm when it sees a *pattern*: many failures from the same source in a short window (rule 5712), which it then classifies as a deliberate brute-force attempt (rule 5763). This is the core idea behind detection engineering — turning ordinary log noise into a small number of meaningful, actionable alerts so an analyst is not overwhelmed.

6. Talking Points (Interview Defense)

- **Why brute-force detection matters:** SSH is one of the most common entry points attackers target on internet-facing Linux hosts. Catching the guessing phase lets a defender respond before the account is compromised.

- **Why the rule chain, not just one rule:** Alerting on every failed login would bury an analyst in false positives. Escalating only when failures cluster keeps the signal high and the noise low.
- **How I would extend this:** add an active response to auto-block the source IP after rule 5763 fires, and forward the alert to a ticketing or notification channel so it is actioned, not just logged.

7. Evidence Index

evidence/A2-S06-wazuh-agent-active-dashboard.png	Agent enrolled and active
evidence/A2-S07-ssh-bruteforce-detection-5763.png	Brute-force detection firing (rule 5763)